WEEKLY LITERATURE REVIEW REPORT

Project: Post-Disaster Assessment using ML with Semantic Communication

Timeline: Week 1

1. KEY PAPER SUMMARY

• Paper Title: A Machine Learning Approach for Post-Disaster Data Curation

• Core Methodology Studied: The authors utilized a hybrid computer vision and natural language processing (NLP) framework to automate data curation. They paired pre-trained deep learning recognition models (such as Amazon Rekognition) to extract secondary visual labels, then fed these into a Word-Embedding model (GloVe) and a Term Frequency-Inverse Document Frequency (TF-IDF) query engine to calculate similarity scores across a custom structural ontology.

• Relevance to Our Project: This paper addresses the "Data Rich Information Poor" (DRIP) problem common in disaster response, where unorganized massive image storage delays forensic assessments. It directly informs our pipeline by showing how automated textual label extraction and word vector mapping can reduce the reliance on tedious manual bounding-box indexing. This fits perfectly with a semantic communication framework, which relies on compressing raw pixel observations into high-level descriptive intents.

2. IN-TEXT CITATION EXAMPLE

"As historical data archives enter a big data era, automated indexing methods must replace manual annotation frameworks to prevent processing delays during structural vulnerability profiling and forensic hazard analysis [1]."

[1] S. H. Ro, Y. Li, and J. Gong, "A Machine Learning Approach for Post-Disaster Data Curation," International Journal of Disaster Risk Reduction, vol. 104, p. 104374, Apr. 2024.

2. KEY PAPER SUMMARY

• Paper Title: FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding

• Core Methodology Studied: The authors presented a high-resolution Unmanned Aerial Vehicle (UAV) imagery dataset benchmarked against deep convolutional neural networks to evaluate baseline models across three simultaneous computer vision tasks: Image Classification (scene-level attributes), Pixel-Wise Semantic Segmentation (isolating flooded vs. non-flooded structures), and Visual Question Answering (VQA).

• Relevance to Our Project: Traditional disaster remote sensing relies heavily on satellite imagery, which introduces high latency, expensive deployments, and severe cloud/smoke interference. FloodNet demonstrates how low-altitude UAV flights bypass these limitations to capture crisp structural data. For our semantic communication project, the paper's focus on pixel-wise semantic segmentation vs. macro classification provides the mathematical justification for task-oriented networking: transmitting dense pixel labels over crippled emergency networks is highly inefficient, establishing the clear necessity for compact semantic classification pipelines at the edge.

2. IN-TEXT CITATION EXAMPLE

"While satellite observations suffer from cloud obscuration and limited spatial resolution, low-altitude Unmanned Aerial Vehicles (UAVs) present an efficient tactical alternative for extracting high-fidelity structural data during sudden flood and hurricane emergencies [2]."

REFERENCES (IEEE Format)

[2] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. Murphy, "FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, pp. 3277-3286.